

Lab experiment-6

Experiment 6: Vacuum Cleaner Problem

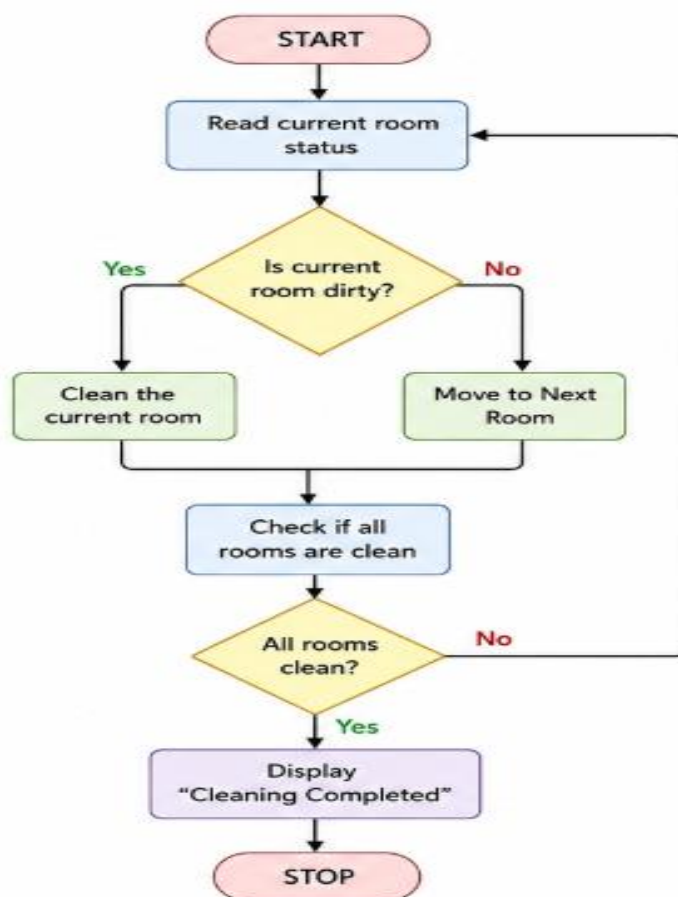
Aim

To implement the Vacuum Cleaner problem using Python.

Objective

To clean all dirty rooms by allowing the vacuum agent to move intelligently between rooms.

Flow chart



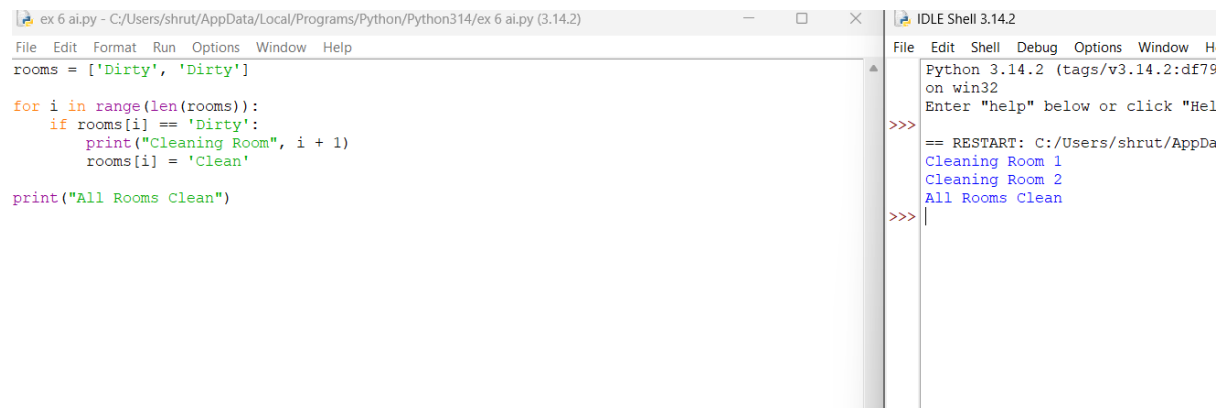
Algorithm

1. Start the vacuum in one room.
2. Check whether the current room is dirty.
3. If dirty, clean it.
4. Move to the next room.
5. Repeat until all rooms are clean.
6. Stop the program.

Code

```
rooms = ['Dirty', 'Dirty']  
for i in range(len(rooms)):  
    if rooms[i] == 'Dirty':  
        print("Cleaning Room", i + 1)  
        rooms[i] = 'Clean'
```

`print("All Rooms Clean")`**Output**

A screenshot of a Python IDE window titled 'ex 6 ai.py - C:/Users/shrut/AppData/Local/Programs/Python/Python314/ex 6 ai.py (3.14.2)'. The code editor on the left contains the following Python code:

```
rooms = ['Dirty', 'Dirty']  
  
for i in range(len(rooms)):  
    if rooms[i] == 'Dirty':  
        print("Cleaning Room", i + 1)  
        rooms[i] = 'Clean'  
  
print("All Rooms Clean")
```

The output console on the right, titled 'IDLE Shell 3.14.2', shows the execution output:

```
Python 3.14.2 (tags/v3.14.2:df79  
on win32  
Enter "help" below or click "Hel  
>>>  
== RESTART: C:/Users/shrut/AppDa  
Cleaning Room 1  
Cleaning Room 2  
All Rooms Clean  
>>>|
```

Result

The vacuum agent cleaned all dirty rooms successfully.

Conclusion

The Vacuum Cleaner agent completed the cleaning process efficiently.